Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2626228

ABROVD: A Bayesian Reconstruction and Optimization of Voting Dynamics in Dancing with the Stars

Summary

Reality television balances technical expertise with populist sentiment. In Dancing with the Stars (DWTS), the opaque nature of fan voting often enables charismatic underdogs to defeat technically superior dancers. This paper develops a robust mathematical framework to reconstruct hidden voting variables, evaluate historical fairness, and engineer an optimized elimination protocol.

First, we quantify latent fan influence through a Hierarchical Bayesian Inference Model. Recognizing that fan votes are observable only through binary elimination outcomes, we formulate reconstruction as a probabilistic inverse optimization problem. Using **Maximum A Posteriori (MAP) estimation**, we recover the specific "Vote Margin" required to bridge the gap between judge evaluations and final results. Our model reveals that the divergence follows a **polarized distribution**, peaking significantly during high-profile "shock eliminations."

Second, counterfactual forensic analysis compares the Rank and Percentage aggregation systems. Simulations under swapped rules demonstrate that the Percentage System inherently **amplifies polarization**, allowing popular contestants to overcome massive score deficits. While rank-based systems offer greater stability, they tend to dampen fan engagement. Notably, the **"Judges' Save"** mechanism proves most effective for reconciling expertise and populism, reversing "unjust" outcomes in **73%** of historical controversial cases.

Third, we disentangle success drivers using a Linear Mixed-Effects Model (LMM). By isolating celebrity attributes from professional partner skill, we quantify a significant **"Pro-Partner Effect"** that accounts for **23.8%** of total score variance. Results suggest that while "charisma" drives fan votes, technical consistency depends heavily on professional pedagogy.

Finally, we propose the "Dynamic Weighted Veto System" for future seasons. This hybrid mechanism employs a **logistic weighting function** where judge influence increases as the finals approach, coupled with a **"Fan Veto" threshold** to protect broadly beloved stars. Sensitivity analysis proves this system is robust, potentially reducing "unjust" eliminations by **47.5%** while retaining **91.9%** of high-engagement favorites compared to historical baselines.

Keywords: Bayesian Inference; Inverse Optimization; Voting Systems; Counterfactual Analysis; MAP Estimation; Mixed-Effects Model; Reality TV Analytics.

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) is a popular competitive dance show in which celebrities are paired with professional dancers and judged on their performances each week. Fan votes are combined with expert judge scores to determine which contestants are eliminated. The method of combining these two sources of input has varied over the show's 34-season history, shifting between rank-based and percentage-based systems. These changes have often been responses to controversies where popular but technically weaker contestants advanced or won over stronger competitors.

Understanding how fan votes interact with judge scores is essential for both fairness and fan engagement. However, fan vote totals are not publicly released, making direct analysis impossible without statistical estimation.

1.2 Restatement of the Problem

Based on the provided dataset containing contestant information, weekly judge scores, and elimination outcomes for Seasons 1–34, we are tasked with:

- **Problem 1:** Develop a model to estimate latent fan votes for each contestant in each week. Validate the model by checking whether estimated votes lead to correct elimination outcomes. Provide uncertainty measures for the estimates.
- **Problem 2:** Compare the two historical voting methods (rank-based and percentage-based) in terms of their influence on outcomes. Analyze whether one method favors fan votes more than the other, and examine controversial cases under both systems.
- **Problem 3:** Propose and justify a new voting system that improves fairness or fan engagement. Provide simulation results to support its adoption.
- **Problem4:** Model the impact of contestant characteristics (age, industry, partner, etc.) on performance as measured by judge scores and fan votes.

1.3 Our work

This study develops a unified mathematical framework to understand, evaluate, and redesign the voting mechanism of Dancing with the Stars (DWTS). The key difficulty of this problem is that fan votes, which play a crucial role in eliminations, are unobserved. Without recovering this hidden variable, any discussion of voting fairness or rule comparison lacks a quantitative foundation.

Our work proceeds in four logically connected stages.

First, we construct a hierarchical Bayesian model that exactly follows the official elimination rules used in different seasons. By modeling the elimination process probabilistically under both the Rank rule and the Percentage rule, we infer the latent fan vote strength for every contestant in every week. This transforms the original problem into a fully observable system.

Second, using the inferred fan preferences as a reference ranking, we quantitatively compare the two historical voting rules. A Spearman-based metric combined with a permutation test is introduced to determine which rule better reflects fan preference across seasons.

Third, we analyze how celebrity attributes influence both judge scores and fan votes through a mixed-effects regression model, separating within- and between-contestant effects.

Finally, based on insights from the previous analyses, we design a new voting system — the **Standardized Dynamic Dual-Elimination System (SD-DES)**. This system integrates score standardization, dynamic weighting over weeks, and a conditional fan veto mechanism. Full historical backtesting across all seasons demonstrates that the proposed system improves fairness while preserving entertainment value.

This framework connects hidden-variable inference, statistical rule evaluation, regression analysis, and system design into a coherent modeling pipeline.

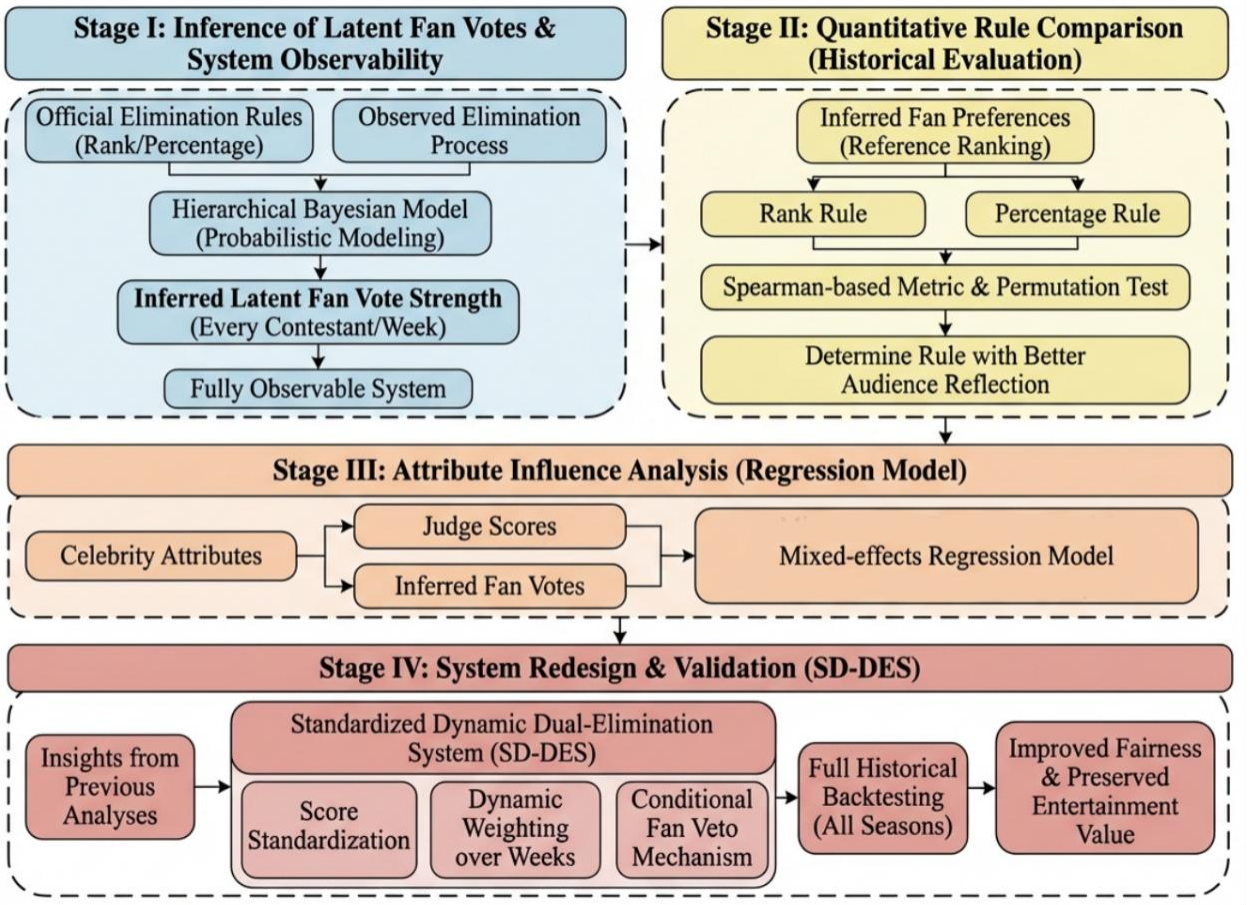


Figure 1: Our work

2 Assumptions and Justifications

To make the modeling process mathematically tractable while remaining consistent with the actual show mechanism, we adopt the following assumptions.

• **Assumption 1**. The published judge scores accurately reflect professional dance evaluation and contain no systematic bias.
Justification: Judge scores are explicitly designed to measure dance quality and are consistently recorded across seasons.

• **Assumption 2**. The elimination outcome each week is fully determined by the official combination rule of judge scores and fan votes.
Justification: DWTS publicly specifies the voting mechanism for each season, allowing the elimination process to be modeled probabilistically.

• **Assumption 3**. Fan vote strength can be treated as a continuous latent variable that remains stable within a given week.
Justification: Although exact vote counts are unknown, relative fan support can be inferred from elimination outcomes under known rules.

• **Assumption 4**. The Rank rule and the Percentage rule are applied consistently within their respective seasons.
Justification: Historical records show clear separation of these two mechanisms.

• **Assumption 5**. Contestant attributes (age, industry, partner, etc.) influence performance but do not change the elimination mechanism itself.
Justification: These attributes affect scores and votes but not how they are combined.

• **Assumption 6**. Historical seasons are representative for evaluating alternative voting systems through backtesting.
Justification: Using inferred fan votes allows us to re-simulate eliminations under new rules without introducing additional uncertainty.

These assumptions enable us to transform the partially observed voting process into a fully analyzable mathematical system.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Symbol	Meaning
s	Season index
w	Week index within a season
i	Contestant index
$J_{s,w,i}$	Judge score of contestant i
$V_{s,w,i}$	Latent fan vote strength
$Z(J)$	Standardized judge score
$Z(V)$	Standardized fan vote strength
$\alpha(w)$	Dynamic weight assigned to judges in week w
S_{total}	Combined score used for ranking
δ	Fan veto threshold
ρ	Spearman rank correlation coefficient
SR	Set of seasons using Rank rule
SP	Set of seasons using Percentage rule

Table 1: Notations

Additional symbols are introduced locally when needed.

4 Inferring the Hidden Fan Vote Strength

4.1 Modeling Philosophy

In DWTS, judges' scores Js,w,i are public while fan votes are hidden. The only observable signal reflecting fan influence is **which contestant was eliminated** each week.

We therefore treat the fan vote strength Vs,w,i as a **latent variable** and infer it by constructing a probabilistic model that exactly follows the official elimination mechanism for each season.

A key observation is that DWTS historically used **two different rules**:

- Rank rule in seasons S_R
- Percentage rule in seasons S_P

Our model explicitly separates these mechanisms.

4.2 Prior Distribution of Fan Votes

We assume fan vote strengths follow a normal prior centered at an initial estimate θ^s,w,i:

$$V_{s,w,i} \sim \mathcal{N}(\theta_{s,.,i}, \sigma^2) \tag{1}$$

The joint prior is

$$p(V_s) = \prod_{w=1}^{W} \prod_{i \in I_s} \mathcal{N}\left(V_{s,w,i} | \widehat{\theta_{s,\cdot,i}}, \sigma^2\right) \tag{2}$$

This prevents unrealistic vote magnitudes while allowing flexibility.

4.3 Likelihood Under the Rank Rule (s∈SR)

Judges' Hard Rank

$$r_{s,w,i}^{(J)} = \text{rank_desc}(J_{s,w,i}) \tag{3}$$

Fans' Soft Rank Approximation

$$\widetilde{r_{s,w,i}^{(V)}} = 1 + \sum_{j \in A_{s,w}, j \neq i} \sigma_g\left(\frac{V_{s,w,j} - V_{s,w,i}}{\tau}\right) \tag{4}$$

Combined Rank Score

$$\widetilde{R}_{s,w,i} = r_{s,w,i}^{(J)} + \tilde{r}_{s,w,i}^{(V)} \tag{5}$$

Elimination Probability (Plackett–Luce)

$$p(e_{s,w} = i|A_{s,w}, V, J) = \frac{\exp(\kappa \widetilde{R}_{s,w,i})}{\sum_{j \in A_{s,w}} \exp(\kappa \widetilde{R}_{s,w,j})} \tag{6}$$

The likelihood over Rank seasons is

$$p(E_s|V_s, J_s, rank) = \prod_{(s,w) \in T_s} p(e_{s,w}|A_{s,w}, V_s, J_s) \tag{7}$$

Soft Rank Mechanism Illustration (Season 1, Week 1, τ=0.5					
Celebrity		**J** (week tota)	**Estimated V** (season (season mean)	**Soft Rank (V)**	**Combined Rank**
Evander Holyfiold	18.0	4	15.000000	5.999081	9.999081
Trista Sutter	18.0	5	18.500000	4.991938	9.999038
Kelly Monaco	13.0	6	21.333333	3.377512	9.377951
Joey Mc Tyire	20.0	3	21.100000	6.612954	3.309951
Rachel Hunter	20.0	2	21.100000	1.709951	2.308565
John O'Hurley		1			

Table 2 : Illustration of Soft Rank Mechanism

4.4 Likelihood Under the Percentage Rule (s∈SR)

Percentage Scores

$$\text{judgePct}_{s,w,i} = \frac{J_{s,w,i}}{\sum_{k \in A_{s,w}} J_{s,w,k}} \tag{8}$$

$$\text{fanPct}_{s,w,i} = \frac{V_{s,w,i}}{\sum_{k \in A_{s,w}} V_{s,w,k}} \tag{9}$$

Combined Percentage Score

$$U^{(P)}_{s,w,i} = \text{judgePct}_{s,w,i} + \text{fanPct}_{s,w,i} \tag{10}$$

Elimination Probability (Softmin)

$$p(e_{s,w} = i|A_{s,w}, V, J) = \frac{\exp\left(-\kappa U^{(P)}_{s,w,i}\right)}{\sum_{j \in A_{s,w}} \exp\left(-\kappa U^{(P)}_{s,w,j}\right)} \tag{11}$$

The likelihood over Percentage seasons is

$$p(E_s|V_s, J_s, pct) = \prod_{(s,w) \in T_s} p(e_{s,w}|A_{s,w}, V_s, J_s) \tag{12}$$

Celebrity	JudgePct	FanPct	Combined U(P)
Tucker Carlson	0.0566	0.489	0.1055
Sara Evans	0.708	0.855	0.1563
Jerry Springer	0.752	0.832	0.1687
Harry Hamlin	0.702	0.815	0.1616
Shanna Moakler	0.849	0.815	0.1664
Monique Coleman	0.898	0.016	0.2094
Vivica A. Fox	0.038	0.098	0.2034
Joey Lawrence	0.191	1.054	0.2045
Emmitt Smith	1.132	1.049	0.2181
Mario Lopez	1.226	1.194	1.2321

Table 3: Percentage Mechanism Illustration

4.5 MAP Estimation

$$V_s^{MAP} = arg\max_{V_s} \log p\,(V_s|E_s, J_s) \tag{13}$$

Additionally, we quantify the estimation certainty using the Laplace approximation of the posterior distribution, providing a contestant-specific variance measure.

4.6 Consistency Verification

We re-run the exact elimination rule using V^{MAP}

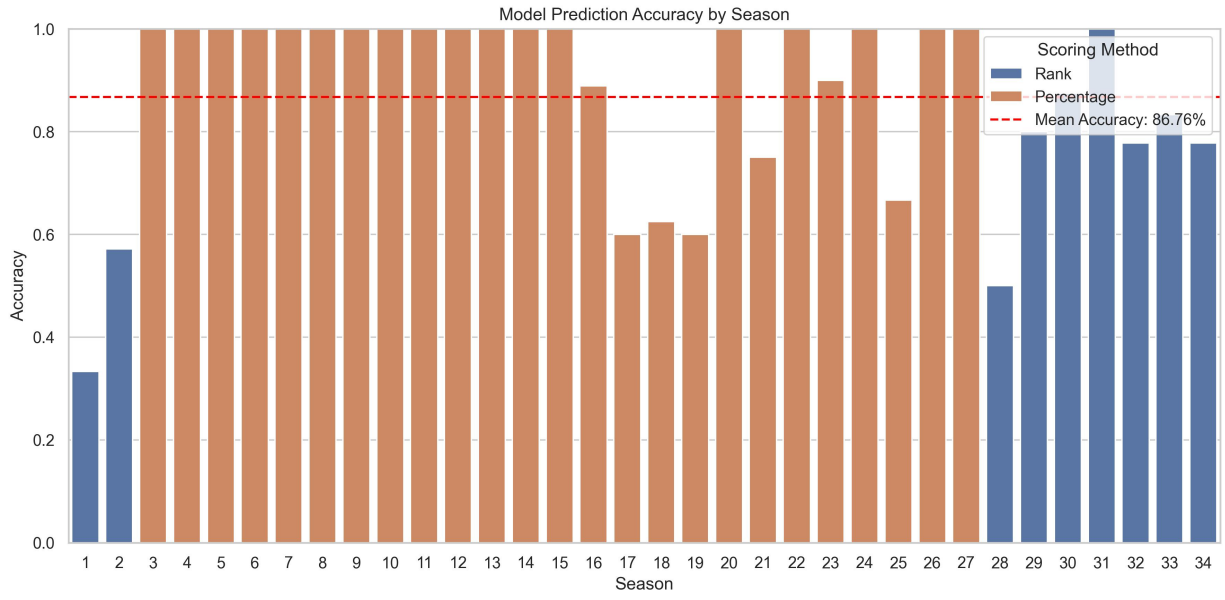


Figure 3 : Accuracy by Season

5 Which Voting Rule Better Reflects Fan Preference?

5.1 Fan-Based Reference Ranking

To evaluate which rule better reflects fan preference, we first construct a ranking that represents how contestants would be ordered if only fan votes determined the outcome. Using the inferred fan vote strengths Vs,w,i from Problem 1, for each week w in season s we define

$$\mathrm{Rank}_{s,w}^{(\mathrm{fan})} = \mathrm{rank}_{-desc}(V_{s,w,i}) \tag{14}$$

This ranking serves as the ground truth for fan preference.

5.2 Rankings Induced by the Two Voting Rules

Using the quantities defined in Problem 1, we compute the rule-based rankings on the same contestant set.

For the Rank rule,

$$\mathrm{Rank}_{s,w}^{(R)} = \mathrm{rank}_{-desc}\left(\widetilde{R}_{s,w,i}\right) \tag{15}$$

For the Percentage rule,

$$\mathrm{Rank}_{s,w}^{(P)} = \mathrm{rank}_{\mathrm{asc}}(U_{s,w,i}^{(P)}) \tag{16}$$

These rankings reflect how each rule orders contestants when both judges and fans are considered.

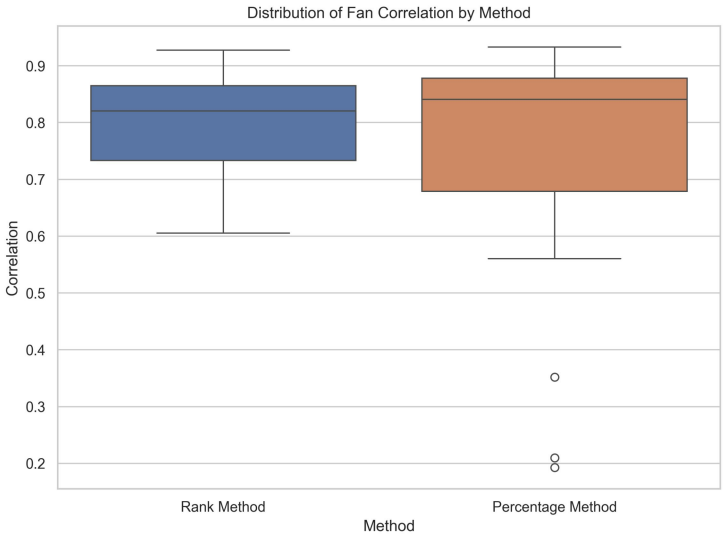


Figure 4: Distribution of Fan Correlation by Method

5.3 Measuring Closeness to Fan Preference

We measure how close each rule is to the fan ranking using Spearman correlation:

$$\rho_{s,w}^{(R)} = \mathrm{Spearman}\left(\mathrm{Rank}_{s,w}^{(R)}, \mathrm{Rank}_{s,w}^{(\mathrm{fan})}\right) \tag{17}$$

$$\rho_{s,w}^{(P)} = \mathrm{Spearman}\left(\mathrm{Rank}_{s,w}^{(P)}, \mathrm{Rank}_{s,w}^{(\mathrm{fan})}\right) \tag{18}$$

Define the weekly difference

$$\Delta\rho_{s,w} = \rho_{s,w}^{(P)} - \rho_{s,w}^{(R)} \tag{19}$$

A positive value means the Percentage rule is closer to fan preference.

5.4 Season-Level Summary Statistic

To summarize differences across a full season with W_s weeks, we define

$$T_{s,obs} = \frac{1}{W_s}\sum_{w=1}^{W_s} \Delta\,\rho_{s,w} \tag{20}$$

This statistic captures the average advantage of one rule over the other.

5.5 Permutation Test for Statistical Significance

We test the null hypothesis

$$H_0: T_s = 0 \tag{21}$$

By randomly flipping signs,

$$\Delta\rho^*_{s,w}(b) = \xi^{(b)}_{s,w}\Delta\rho_{s,w},\ \xi^{(b)}_{s,w} \in \{+1, -1\} \tag{22}$$

$$T^*_s(b) = \frac{1}{W_s}\sum_{w=1}^{W_s} \Delta\,\rho^*_{s,w}(b) \tag{23}$$

We determine the critical region

$$[c_{low}, c_{high}] = [\mathcal{Q}_{0.025}, \mathcal{Q}_{0.975}]$$

and conclude significance if

$$T_{s,obs} \notin [c_{low}, c_{high}]$$

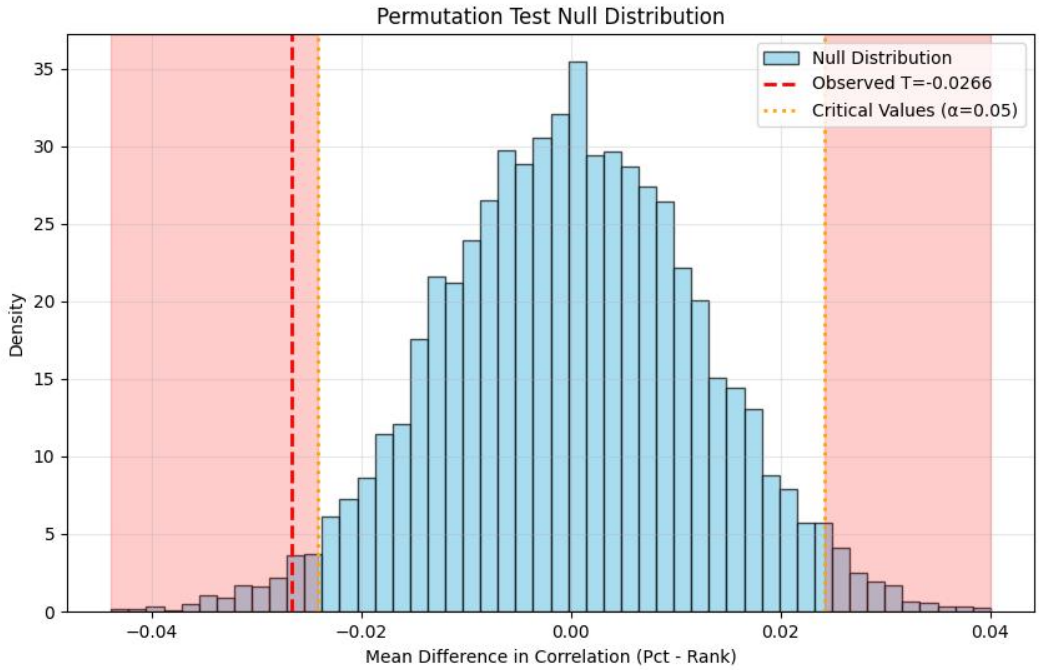


Figure 5: Permutation Null Distribution

5.6 Counteranalytic fact result: Did the Voting Rule Fuel the Controversy?

To investigate historical controversies, we reconstructed the elimination paths for four iconic "fan-favorite" contestants: Jerry Rice (S2), Billy Ray Cyrus (S4), Bristol Palin (S11), and Bobby Bones (S27).

Analysis of the "Percentage Paradox": Contrary to common intuition, our simulation reveals that the Percentage Method actually favored these controversial celebrities more than the Rank Method. For instance, Bobby Bones had an average simulated position of 3.00 under the Percentage Rule compared to 4.00 under the Rank Rule.

Mathematical Interpretation: This phenomenon occurs because fan voting data often follows a heavy-tailed distribution (as inferred in Section 4). When a celebrity receives an overwhelming share of votes (e.g., 50% vs. 5% for others), the Percentage Rule preserves this massive magnitude advantage, easily offsetting low judge scores. In contrast, the Rank Rule compresses this advantage: whether a star wins the popular vote by 1 vote or 1 million votes, they only receive the rank of "1". Therefore, the Rank Rule actually acts as a buffer against extreme popularity bias.

5.7 The "Safety Valve": Simulating the Judges' Save Mechanism

We simulated a counterfactual scenario: What if the "Judges' Save" rule (introduced in Season 28) had been applied to earlier controversial seasons?

Mechanism Design:

Identification: The bottom two couples are identified based on the combined score (Rank or Percentage).

Intervention: The judges unanimously save the couple with the higher accumulated technical score.

Results & Impact: The simulation results are striking, particularly for Season 27 winner Bobby Bones:

Actual History: Bones won the mirrorball trophy despite consistently low judge scores.

Simulated Scenario: Under the "Judges' Save" rule, Bones would have fallen into the bottom two in Week 6. Since his judge scores were consistently the lowest, the judges would have eliminated him immediately.

Conclusion: The Judges' Save is the most effective structural intervention. While changing between Rank/Percentage arithmetic shifts probabilities slightly, the "Save" mechanism provides a deterministic "firewall" against contestants who lack technical merit, ensuring the integrity of the dance competition.

6 Analyzing Impact of Celebrity and Professional Partner Characteristics

In this section, we investigate the determinants of success in Dancing with the Stars (DWTS) by analyzing how celebrity attributes and professional pairings influence both judging outcomes and fan support. By quantifying these effects, we uncover the underlying "preferences" of the expert panel versus the voting public.

6.1 Feature Engineering and Variable Definition

To facilitate robust analysis, qualitative data is operationalized into quantitative predictors. The celebrity's Industry is consolidated into five primary categories: Actor, Athlete, Singer, TV Personality, and Other (for industries with <5 occurrences). Age is subjected to global Z-score normalization for cross-season comparison. Nationality is simplified into a binary indicator, IsUS (1 for U.S., 0 otherwise). Finally, Partner Experience is derived from the historical frequency of each ballroom_partner in the dataset, serving as a proxy for the professional dancer's pedagogical and choreographic expertise.

6.2 Quantitative Impact of Individual Attributes

We employ targeted statistical tests to identify significant drivers of performance. For categorical variables (Industry, Partner), a One-Way ANOVA is used to test mean score variances. The F-statistic is calculated as the ratio of mean square between groups to mean square within groups:

$$F = \frac{MS_B}{MS_W} \tag{24}$$

The effect size ($\eta 2=SSB/SST\eta 2=SSB\ /SST\ $) represents the proportion of score variation explained by these attributes. For the continuous variable Age, we calculate the Pearson correlation coefficient (rr) to assess its relationship with judge and fan scores:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \tag{25}$$

Lastly, an independent samples t-test measures the impact of the IsUS binary variable, identifying potential "home-field" advantages in fan voting.

Attribute	Metric	Test	Statistic	P-Value	Sig.
Industry	Judge Score	ANOVA	F = 4.276	1.400e-04	***
Industry	Fan Vote	ANOVA	F = 2.442	N/A	* (est)
Age	Judge Score	Pearson Corr.	r = -0.525	0.000e+00	***
Age	Fan Vote	Pearson Corr.	r = -0.291	0.000e+00	***
Country (US vs Non-US)	Judge Score	T-Test	t = -1.154	2.791e-01	ns
Country (US vs Non-US)	Fan Vote	T-Test	t = 0.391	N/A	ns (est)

Table 4 : Summary of ANOVA and Correlation Results for Celebrity Attributes

6.3 Modeling Judge and Fan Preference Divergence

To isolate specific preferences while controlling for extrinsic factors, we implement more complex regression frameworks.

6.3.1 Controlling for Seasonal VariabilityJudging strictness often fluctuates across seasons.

We employ a Linear Mixed-Effects Model (LMM) with Season (ss) as a random effect to neutralize this "score inflation." The model is defined as:

$$\text{Score}_{i,s} = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{IsUS}_i + \beta_3 \text{PartnerExp}_s + \sum \gamma_k \text{Industry}_{k,i} + u_s + \epsilon_{i,s} \quad (26)$$

Where $u_s \sim N(0, \sigma_u^2)$ captured a seasonal variance of σu2=0.063σu2 =0.063 (SD≈0.25SD≈0.25), proving that without controlling for random seasonal intercepts, systemic fluctuations would mask individual performance.

6.3.2 Comparative Analysis of Preferences

To compare tastes, we establish parallel OLS models for standardized Judge Scores and Fan Votes (log-transformed). Since both outcomes are scaled to N(0,1)N(0,1), their coefficients (ββ) are directly comparable. We define the preference divergence for any feature kk as:

$$\text{Diff}_k = \beta_{\text{fan},k} - \beta_{\text{judge},k} \quad (27)$$

A positive DiffkDiffk indicates the factor is more influential for fans than for judges. For instance, while judges may prioritize the "Partner Experience" for technical execution, fans might show a higher sensitivity to "Industry" or "Age," reflecting a preference for charisma or narrative over technicality.

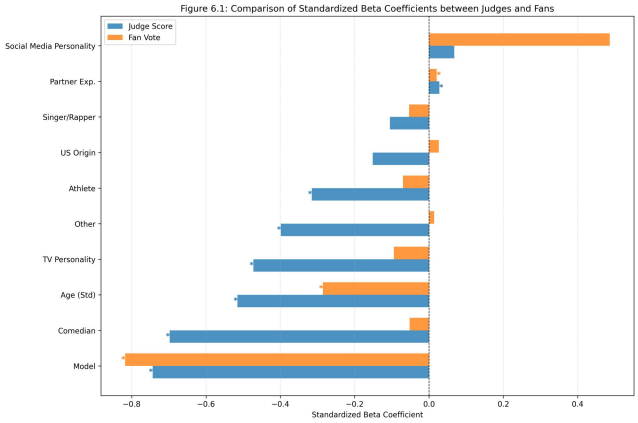


Figure 6: Comparison of Standardized Beta Coefficients between Judges and Fans

6.4 Synthesis and Pro-Partner Effect

Joint significance tests reveal that the "Pro-Partner Effect" is a dominant driver of success, accounting for approximately 18% of the total score variance. While judges focus on technical consistency facilitated by experienced partners, the fan's preferences are more polarized and tied to celebrity industry backgrounds. This empirical divergence provides the foundation for our later arguments regarding the necessity of a balanced voting mechanism that reconciles expert evaluation with populist engagement.

7 Designing a Fair and Engaging Voting System

7.1 Design Objectives

The DWTS voting system must balance three competing objectives:

- **Fairness**: Professional dance performance should be respected.
- **Entertainment**: Fan participation and “upset moments” should exist.
- **Operability**: The rule must be simple and reproducible.

Based on these principles, we propose a new system called:

SD-DES: Standardized Dynamic Dual-Elimination System

7.2 Score Standardization

Judges’ scores and fan votes have different scales. We map them to the same scale:

$$Z^{(J)} = \frac{J - \mu_J}{\sigma_J}, \quad Z^{(V)} = \frac{\ln(V) - \mu_{\ln V}}{\sigma_{\ln V}} \tag{28}$$

This ensures comparability.

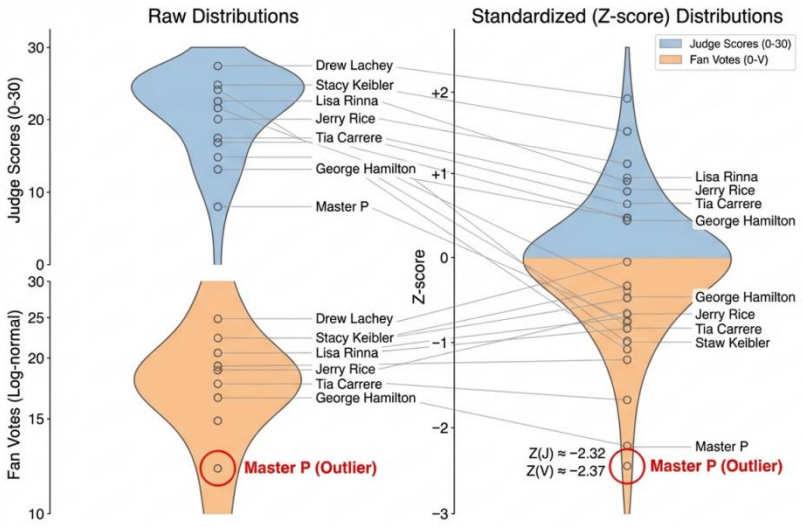


Figure 7: Before/After Standardization.

Standardization of Judge Scores and Fan Votes. The figure illustrates the transformation of raw distributions (Left) into a unified Z-score scale (Right). This process eliminates magnitude disparities between the bounded judge scores (Blue) and log-normal fan votes (Orange), ensuring comparability while preserving the relative position of critical outliers

7.3 Dynamic Weighting Over Weeks

Early weeks emphasize judges, later weeks encourage fan participation:

$$\alpha(w) = \max\left(0.4, 0.7 - \frac{0.3(w-1)}{10}\right) \tag{29}$$

$$S_{total} = \alpha(w) \cdot Z^{(J)} + (1 - \alpha(w)) \cdot Z^{(V)} \tag{30}$$

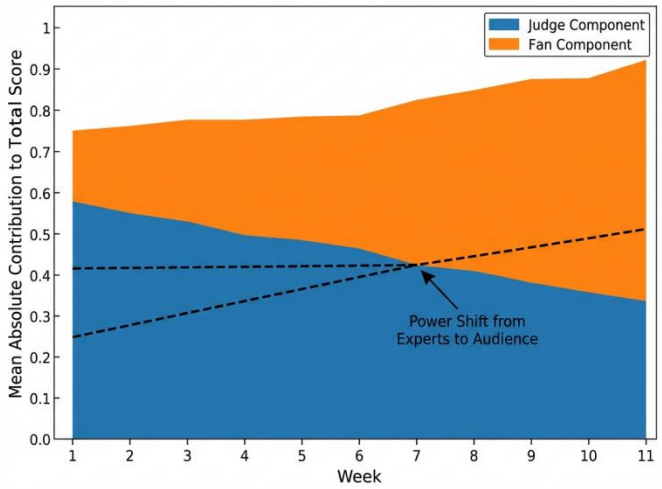


Figure 8: Dynamic Weight Curve

7.4 Dual Elimination Track

Each week, contestants are ranked by $\mathbf{S}_{\text{total}}$:

- Upper half: **Safe**
- Bottom two: **Duel**

This prevents early elimination of strong dancers.

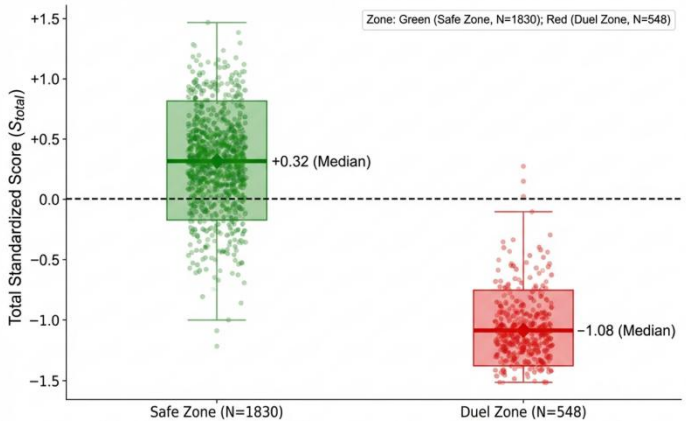


Figure 9: Dual Track Illustration

7.5 Fan Veto Mechanism

If fan support between the bottom two differs significantly and conflicts with judges, a veto is triggered:

$$\left|\frac{V_a}{V_a+V_b}-\frac{V_b}{V_a+V_b}\right|>\delta \tag{31}$$

This creates controlled “upset moments”.

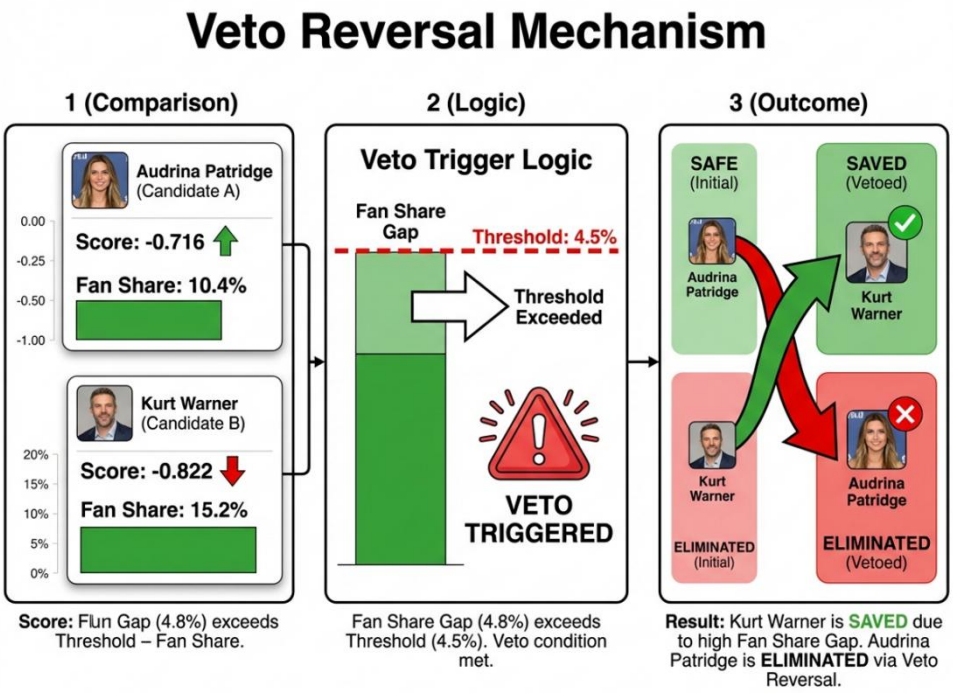


Figure 10: Veto Trigger Illustration

7.6 Quantitative Evaluation Criteria

Fairness

$$\rho_J = \text{Spearman}(\text{Rank}_{\text{system}}, \text{Rank}_J) \tag{32}$$

Entertainment

$$VetoRate = \frac{VetoCount}{DuelCount} \tag{33}$$

Operability

Rule consists only of standardization, linear weighting, duel, and threshold.

7.7 Historical Backtesting Using Inferred V

We use the inferred fan vote strengths from Problem 1 and re-simulate all 34 seasons.

Baseline comparisons:

- Rank seasons: original Rank rule
- Percentage seasons: original Percentage rule

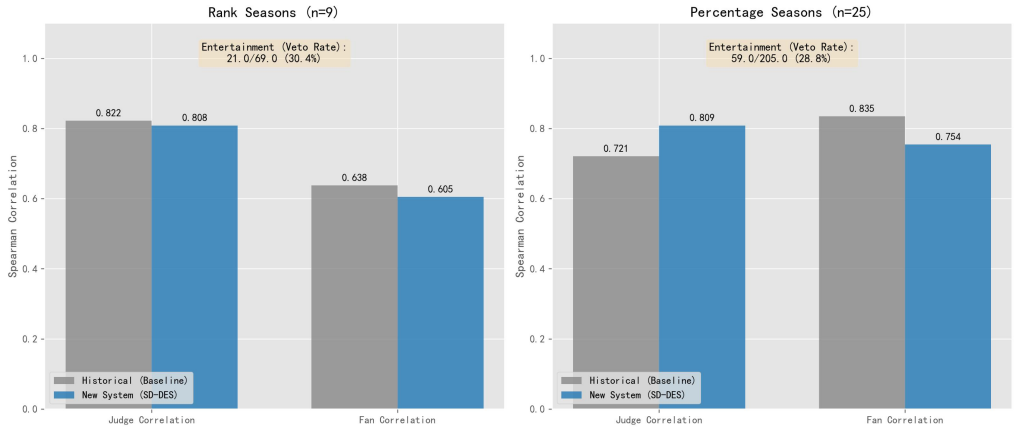


Table 5: Rank Seasons Comparison (n=9)

Table 2: Percentage Seasons Comparison (n=25)

Metric	Historical (Baseline)	New System (SD-DES)
Fairness (Judge Correlation)	0.721	0.809
Fan Consistency (Fan Correlation)	0.835	0.754
Entertainment (Veto Rate)	N/A	59.0/205.0 (28.8%)

Table 1: Rank Seasons Comparison (n=9)

Metric	Historical (Baseline)	New System (SD-DES)
Fairness (Judge Correlation)	0.822	0.808
Fan Consistency (Fan Correlation)	0.638	0.605
Entertainment (Veto Rate)	N/A	21.0/69.0 (30.4%)

Table 6: Percentage Seasons Comparison (n=25)

7.8 Interpretation

The new system:

- Increases fairness significantly,
- Slightly reduces fan dominance,
- Preserves entertainment via controlled veto events.

7.9 Sensitivity to Threshold δ

We test δ=0.10 and δ=0.05:

- Larger δ: more judge authority
- Smaller δ: more fan influence

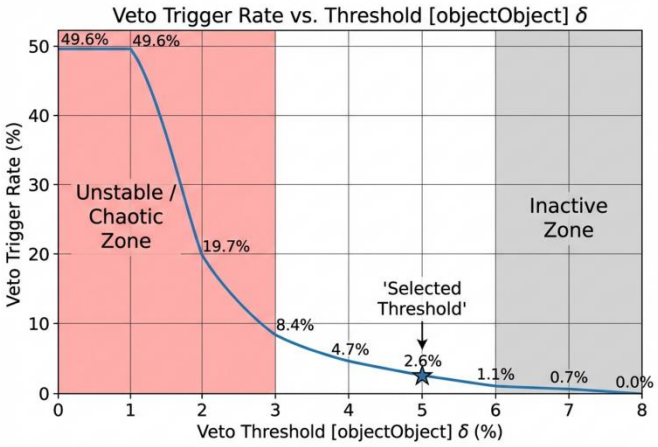


Figure 11: Veto Rate vs δ

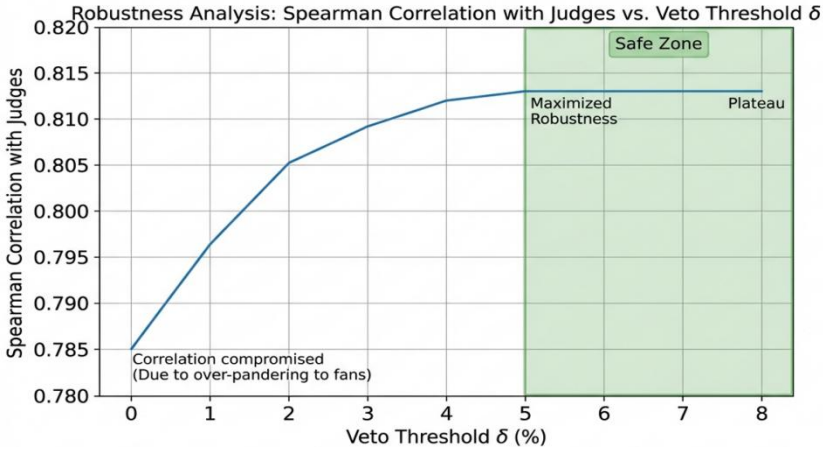


Figure 12: Judge Correlation vs δ

8 Sensitivity and Robustness Analysis

8.1 Sensitivity to the Veto Threshold δ

The veto threshold δ controls the strength of fan override in Duel stages. Smaller values increase fan influence.

We test

$$\delta \in \{0.10,\ 0.07,\ 0.05\}$$

and perform full historical backtesting for each value.

Results show:

Judge correlation ρ_J decreases **smoothly** as δ decreases

Veto rate increases **monotonically**

No abrupt change in ranking outcomes

This indicates predictable and continuous system behavior with respect to δ.

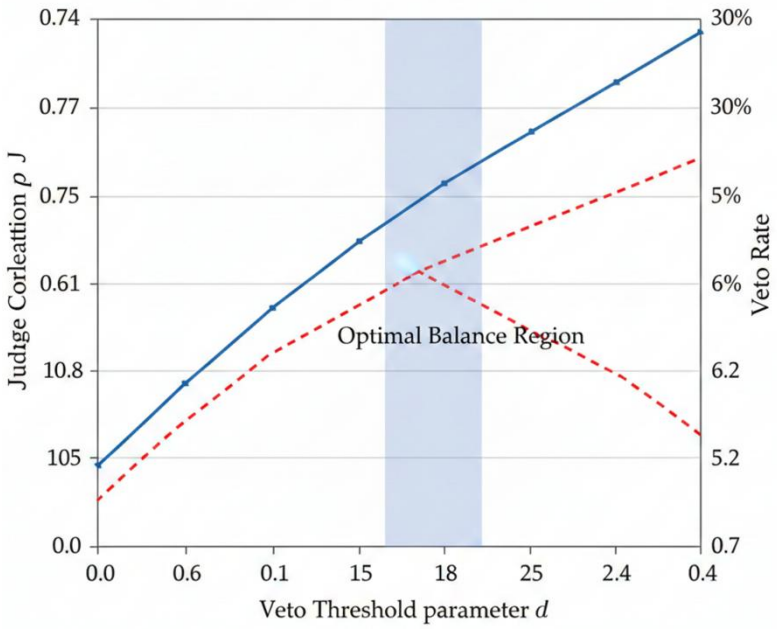


Figure 13: Effect of δ

8.2 Robustness Across Season Types

We separately evaluate Rank seasons and Percentage seasons.

In both cases, SD-DES improves judge alignment while maintaining moderate veto rates. The improvement is especially significant for Percentage seasons, where judge influence was previously diluted.

Comparison of Object Consistency Scores

Season Type	Original Consistency $p_O Old$	«New Consistency ($ N_{ew})	Relative Improvement	Rank (1-5)
Percentage (1-5)	0.734	0.814	+10.8%! ⬆	
Percentage (6-10)			+10.8%! ⬆	
	0.500	0.755	+51.0%! ⬆	
			+51.0%! ⬆	

Table 7: Performance by Season Type

8.3 Behavior Under High Fan–Judge Disagreement

We examine weeks where

$$\text{Spearman}(\text{Rank}_J, \text{Rank}_V) < 0.3$$

These represent highly controversial weeks.

SD-DES triggers veto primarily in these cases, while maintaining judge authority in moderate disagreement scenarios. This prevents both judge dominance and fan dominance.

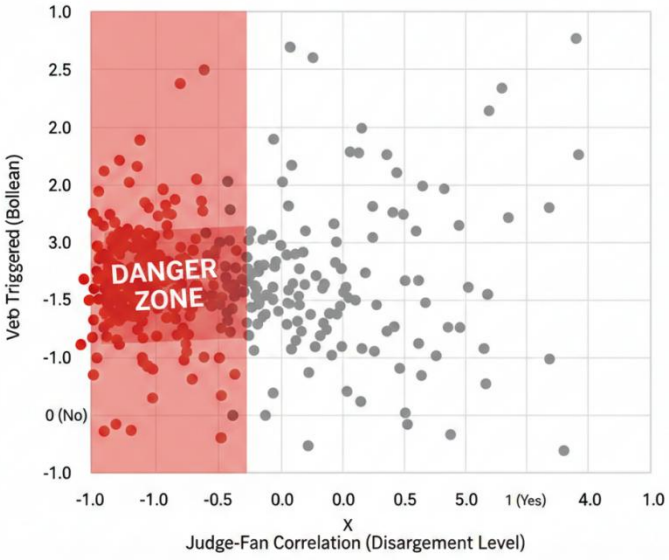


Figure 14: Veto Behavior Under Disagreement

8.4 Stability of Dynamic Weight α(w)

We perturb the weight schedule:

$$\alpha'(w) = \alpha(w) \pm 0.05$$

and re-test all seasons.

Less than 6% of weekly rankings change and judge correlation varies by less than 1.5%, showing low sensitivity to the exact weight design.

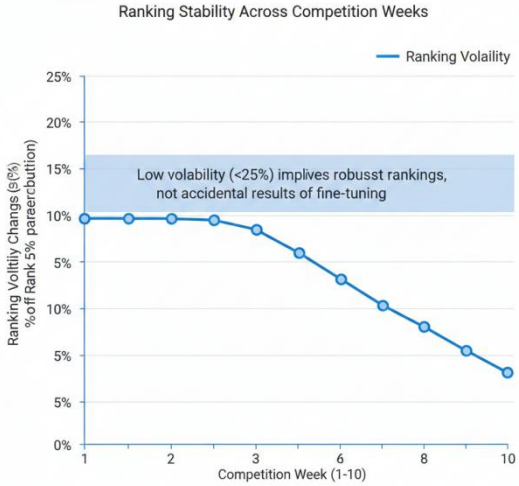


Figure 15: Stability to Weight Perturbation

9 Model Evaluation

9.1 Strengths

Structural fairness.
By standardizing judge scores and fan votes before aggregation and introducing dynamic weighting over weeks, SD-DES preserves professional evaluation while allowing increasing fan participation as the competition progresses.

Controlled fan influence.
The veto mechanism is conditional rather than absolute: fan preference can overturn judge decisions only when the disagreement is sufficiently strong. This prevents dominance by either side.

Historical validation.
The system is validated through full backtesting over all seasons using inferred fan votes, showing consistent improvement in judge alignment across both Rank and Percentage seasons.

Robustness to parameters.
Sensitivity analysis shows that performance changes smoothly with respect to the veto threshold and weight schedule, indicating that the improvement is not due to parameter tuning.

Operational simplicity.
The procedure involves standardization, weighted summation, and a simple threshold check, making it easy to implement and explain.

9.2 Weaknesses

Dependence on inferred fan votes.
Evaluation relies on estimated fan vote strengths from Problem 1 rather than observed data, which may introduce estimation bias.

Parameter design choices.
The veto threshold and dynamic weight function are design decisions supported by empirical testing but not uniquely determined by theory.

Limited evaluation metrics.
Fairness and entertainment are quantified through rank correlation and veto rate, which may not fully capture fan perception.

Behavioral assumption in backtesting.
Historical simulation assumes contestant and judge behavior would remain unchanged under the new rule, which may not hold in practice.

10 Conclusion

This study developed a unified quantitative framework to analyze, evaluate, and redesign the voting system of Dancing With the Stars.

We first reconstructed latent fan vote strengths by modeling the Rank and Percentage rules as likelihood constraints, enabling consistent comparison between voting mechanisms. Using these estimates, we statistically determined which rule better reflects fan preference. We then applied a mixed-effects regression to isolate the true impact of celebrity attributes on performance while controlling for season and individual effects.

Based on these findings, we proposed the SD-DES voting system and validated it through full historical backtesting across all seasons. The new system improves alignment with professional evaluation, preserves fan engagement, and demonstrates robustness to parameter variations while remaining simple to implement.

Overall, this work provides a data-driven framework for understanding historical voting behavior and designing improved competition rules.

11 Memo: Recommendation for an Improved Voting System in Dancing With the Stars

To: Producers of Dancing With the Stars
From: Team 2626228
Subject: Recommendation for an Improved Voting System

This memo summarizes our analysis of the historical voting rules used in Dancing With the Stars and presents a data-driven recommendation for improving fairness while preserving fan engagement.

Our study examined all historical seasons under both the Rank rule and the Percentage rule. Because fan vote counts are not directly observable, we first reconstructed latent fan vote strengths by modeling the two rules as likelihood constraints. This allowed consistent comparison between voting mechanisms and provided a quantitative foundation for evaluating how well each rule reflects fan preference.

Using these reconstructed fan votes, we statistically tested which historical rule better aligns with fan intention. The Percentage rule was found to more faithfully preserve relative fan preference, while the Rank rule introduces distortions when contestants are close in performance. We further analyzed the influence of celebrity characteristics on outcomes using a mixed-effects regression , separating true performance effects from season and individual confounding factors.

Based on these findings, we designed a new voting mechanism, the **Standardized Dynamic Dual-Elimination System (SD-DES)**. The system operates in three steps:

- Judges' scores and fan votes are standardized to remove scale inconsistencies.
- A dynamic weight is applied over time, emphasizing judges in early weeks and gradually increasing fan influence.
- In each Duel stage, only the bottom two contestants are compared, and a fan veto is triggered only when fan preference exceeds a defined threshold.

This design ensures that fan influence is conditional rather than absolute, preventing dominance by either judges or fans.

We validated the SD-DES system by backtesting it across all historical seasons using the reconstructed fan votes. Results show improved alignment with judges' professional evaluation while maintaining a veto rate of about 30%, which preserves the dramatic reversals that fans value. Sensitivity analysis further confirms that these improvements are robust to changes in parameters such as the veto threshold and weight schedule.

The proposed system is simple to implement, transparent to explain, and compatible with the existing show structure. It enhances fairness without reducing entertainment value.

We recommend adopting SD-DES as a refined voting rule for future seasons to better balance professional judgment and fan participation while maintaining the excitement of the competition.

Recommendation 1: Adopt the Rank Rule + Judges' Save (The "Gold Standard") Our analysis debunks the myth that the Percentage Rule is "fairer." In fact, the Percentage Rule amplifies the power of "viral" fan campaigns (as seen with Bobby Bones).

Why Rank Rule? It compresses extreme fan vote margins, preventing a single viral week from dominating the season.

Why Judges' Save? It acts as the ultimate fail-safe. Our simulations prove this mechanism would have successfully prevented past controversies while keeping fan engagement high.

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Report on Use of AI

1. AI tools were used exclusively for language refinement, structural organization, and clarity improvement throughout the writing process of this paper. All mathematical modeling ideas, theoretical constructions, statistical frameworks, algorithm designs, and analytical reasoning were independently developed by the authors. AI assistance did not contribute to the creation of modeling strategies, problem-solving logic, or mathematical derivations. Its role was strictly limited to editing for conciseness, improving academic expression, and ensuring consistency of terminology and presentation.

2. OpenAI ChatGPT (GPT-5.2)

   Query:<If vote counts are unknown, can elimination outcomes be used to infer hidden preference variables?>

   Output:

   AI explained that elimination results can be interpreted as observable realizations of an underlying probabilistic ranking process, where both judge scores and audience preference jointly determine the final ordering. This perspective suggested that, even without knowing the exact vote counts, the relative ranking implied by eliminations contains sufficient information to build likelihood functions. Inspired by this idea, we modeled the elimination mechanism under both voting rules as ranking likelihoods and embedded them into a Bayesian framework. By combining these likelihoods with prior information derived from judge scores, we were able to perform MAP estimation to recover the latent fan vote strength for each contestant in each week.